

Milestone Two Enhancement One: Software Design/Engineering

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The artifact I selected is an Android inventory application that I originally created in CS 360: Mobile Architecture and Programming. The application was developed in Java using Android Studio and uses SQLite for local data storage. It allows users to create an account, log in, add and remove inventory items, adjust item quantities, and optionally receive an SMS notification when an item's quantity reaches zero.

I selected this artifact for my ePortfolio because it demonstrates software design and engineering skills across a complete mobile application. The project uses Java, Android user-interface components, object-oriented programming, event-driven behavior, RecyclerView, multiple activities, and SQLite database operations. It also demonstrates my ability to modify an existing application while preserving its working functionality.

For this enhancement, I added the ability to edit an existing inventory item. A user can select an item, open an Edit Item screen populated with its current name and quantity, modify those values, and save the changes. The enhancement validates the entered values and updates the existing SQLite record by its unique ID. When the user returns to the inventory screen, the records are reloaded so the updated information is displayed. I also revised the inventory-row layout to provide enough space for the new Edit action and improve the readability of individual inventory records.

The enhancement met the course outcomes I planned to address in Module One. It supports Course Outcome 1 by improving the accuracy and maintainability of inventory information that can be used to support inventory-related decisions. It supports Course Outcome 2 through a clear editing workflow, user-interface changes, validation messages, and visual organization of inventory records. Course Outcome 4 is demonstrated through the use of Java, Android Studio, Android activities and intents, RecyclerView, and SQLite to extend a working

software solution with additional functionality that provides value to the user. I do not have any updates to my planned outcome coverage for this enhancement.

Enhancing the application showed me how a relatively small user-facing feature can require coordinated changes across several parts of an application. The selected inventory record had to be identified, its existing values passed to the Edit Item activity, the user's changes validated, and the correct SQLite record updated. The inventory screen also had to reload the stored data when the user returned. Working through this process strengthened my understanding of how the user interface, application logic, data model, and persistence layer work together.

The main challenge I encountered was incorporating the Edit action without making the existing inventory row difficult to read. Adding another button reduced the space available for longer item names. I addressed this by reorganizing the row so that the item information and quantity controls are separated from the Edit and Delete actions and by adding visual separation between inventory records. This reinforced that a functional change can also require interface changes to maintain usability.

I also used incremental testing while making the enhancement. Testing after changes to the model, database logic, editing activity, and interface helped confirm that the new feature worked without disrupting the existing inventory functions. The completed enhancement gave me additional experience extending an existing codebase and considering how one change affects multiple parts of a software application.